

Identifying Monetary Policy Shocks: a Natural Language Approach

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Presented by Janet Choi

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Overview

- How does monetary policy affect the economy?
⇒ Study the response of economic variables to a monetary policy shock
- Main challenge: Policy responds to & affects economic conditions ← identification
⇒ Goal: Isolate changes in interest rates that occur exogenously

monetary policy shock
- This paper: estimate policy shocks using a combination of **natural language processing** and **machine learning techniques**
 - Builds on the approach of Romer & Romer (2004)
 - Exogenous shocks = difference betw. *actual* and *systemic* movements in the target FFR
 - Systemic changes in the FFR can be estimated using proxies for the Fed's information set
 - Including **textual information** to capture the information set leads to cleaner identification

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Preview of Key Results

- Construct **sentiment indicators** for ~ 300 economic concepts from textual content of FOMC staff documents
- Include sentiment indicators in a **ridge regression** to estimate monetary policy shocks
- Baseline result: around 94% of changes in the target FFR due to systemic shocks, **6% due to monetary policy shock**
- In meetings where the estimated policy shock was large, the FOMC made decisions based on considerations other than the staff's forecasts
- Resulting shock measure can produce impulse responses that are consistent with predictions of standard economic theory

1. Monetary Policy Shocks

2. Estimation Method

3. Informational Content of Sentiment

4. Estimation Results

5. Effects of Monetary Policy Shocks

Monetary Policy Shocks

- Monetary policy shock: changes in monetary policy that are exogenous (i.e. not Fed responses to economic conditions)
- Formulation of the problem:

$$\Delta s_t = f(\Omega_t) + \epsilon_t$$

- s_t : policy instrument
- Ω_t : information set of the Fed
- $f(\Omega_t)$: systematic component of monetary policy $\rightarrow$ endogenous
- ϵ_t : monetary policy shock we want to recover, assumed to be orthogonal to $f(\Omega_t)$

Monetary Policy Shocks

$$\Delta s_t = f(\Omega_t) + \epsilon_t \Rightarrow \text{How can we estimate } \epsilon_t?$$

- Romer & Romer (2004):

$$\Delta i_t = \alpha + \beta i_{t-1} + \gamma \mathbf{X}_t + \epsilon_t^{RR}$$

- i_t : FFR target
- $\mathbf{X}_t$: Greenbook forecasts (numerical predictions) of unemployment, output, inflation
- Residuals $\hat{\epsilon}_t^{RR}$: empirical measure of ϵ_t
- Relies on **two assumptions**:
 - 1) $\mathbf{X}_t$ is a good approximation of Ω_t
 - 2) linear specification approximates the unknown mapping $f(\cdot)$ well

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Monetary Policy Shocks

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$$\Delta i_t = \alpha + \beta i_{t-1} + \gamma \mathbf{X}_t + \epsilon_t^{RR}$$

- Relies on two assumptions:
 - 1) forecasts $\mathbf{X}_t$ are able to capture the content of Ω_t
 - 2) linear specification approximates the unknown mapping $f(\cdot)$ well
- This paper suggests modifying the above with :
additional regressors (**sentiment indicators**) + **nonlinearity** (add quadratic terms)
- Sentiment indicators: extracted from textual information, intended to capture how the Fed assesses economic conditions

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Process FOMC Documents

Retrieve FOMC documents for meetings

- Downloaded from Fed Board of Governors website, available with a 5-year lag
- Resulting dataset: documents for a total of **276 meetings from 1982-2016**

Types of documents included:

- Analysis & US economy outlook: *Greenbook 1, 2* (~ 2010/06), *Tealbook A* (2010/06 ~)
- Economic conditions by Fed District: *Redbook* (~ 1983), *Beigebook* (1983~)

Identify Economic Concepts

- Go through each word in each document: identify *singles*, *doubles*, *triples*
 - *Singles*: individual words
 - *Doubles*, *triples*: 2(3)-word expressions that do not contain stop words/sentence breaks
- Rank singles, doubles, & triples by their frequency over sample period
- Among them, identify **economic concepts**
 - Select only those that are economic concepts (based on authors' judgment) ▶ algorithm
 - e.g. credit, reserves (*singles*), labor market (*double*), commercial real estate (*triple*)
 - Total of 296 economic concepts

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Figure 1: ECONOMIC CONCEPTS MENTIONED FREQUENTLY IN FOMC DOCUMENTS



Construct Sentiment Indicators

Example: constructing the sentiment indicator for '*gdp*'

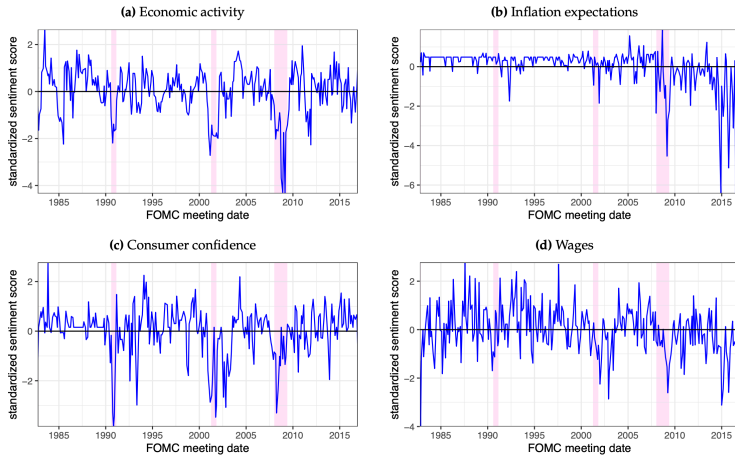
- For each occurrence of '*gdp*' in a document, look at a fixed window of 10 words

“ *gdp* ”
10 words 10 words

- Count the number of **positive & negative words** in the window
 - Based on the Loughran and McDonald (2011) dictionary [▶ example](#)
 - Compute the **sentiment score**: Each positive word +1, negative word -1
 - Repeat for all documents, sum up the score across all documents for a given meeting
 - Repeat for all meetings
- ⇒ End product: **sentiment indicator time series** for each of the 296 economic concepts

Sentiment Indicators

Figure 2: SELECTED SENTIMENT INDICATORS



Empirical Model

$$\Delta i_t = \alpha + \beta i_{t-1} + \Gamma(\tilde{\mathbf{X}}_t, \mathbf{Z}_t) + \epsilon_t^*$$

- Estimate for Oct 1982 - Oct 2008 (avoid $\Delta i_t = 0$), 210 meetings
- i_t : FFR target, $\tilde{\mathbf{X}}_t$: augmented set of numerical forecasts produced by Fed staff
- $\mathbf{Z}_t$: sentiment indicators & 4 lags
- $\Gamma()$: linear quadratic function (nonlinear mapping)
- Want to estimate residuals $\hat{\epsilon}_t^*$, but $\#$ regressors (3,226) $>$ $\#$ obs ($T = 210$)
- Use **ridge regression**: $\min$ 'squared error + $\lambda \sum_j (\text{coefficient}_j)^2$ ' [▶ details](#)
- Optimal λ found using 10-fold cross validation

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Informational Content of Sentiment Indicators

- Main addition to the set of regressors: sentiment indicators extracted from text
- Possible questions:
 - Do sentiment indicators matter for estimating policy shocks (ϵ_t) ?
 - Why should we consider including them as regressors?
- Before examining estimated shock, validation exercise to answer the above

Mean vs. Mode Forecasts

Cochrane (2004): discussion of Romer and Romer (2004)

If Greenbook forecasts incorporate all information available to the FOMC about a variable of interest, it would be **unnecessary** to include additional regressors to estimate a shock that is exogenous to Fed's information set

- Implication: sentiment indicators are *unneeded* in the set of regressors
- Assumption: Greenbook forecasts are the **conditional mean** expectation ($\mathbb{E}[Y_{t+h}|\Omega_t]$)

However, evidence that Greenbook forecasts are best interpreted as modal predictions

⇒ Examples in FOMC meeting transcripts where staff mention forecasts' modal nature

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Mean vs. Mode Forecasts

Examples from the transcripts:

- July 1996 meeting: Michael Prell, director of Research and Statistics

“I would characterize our forecasts over the years as an effort to present a meaningful, **modal forecast of the most likely outcome**. When we felt that there was some skewness to the probability distribution, we tried to identify it. In this instance, as we looked at the recent data, ... a little above our **modal forecast**.”

- March 2005 meeting: Janet Yellen, President of San Francisco Fed

“While the Greenbook expectation of a relatively flat path for bond rates through the end of next year may be a reasonable **modal forecast**, I don’t think the risks here are balanced.”

Forecast Error Predictability

- Even if Greenbook forecasts are modal rather than mean predictions, Cochrane's argument would still apply if $\text{mean} \simeq \text{mode}$ most of the time
- Run **forecast error regressions** to test whether sentiment indicators predict errors in the Greenbook forecasts of the unemployment rate (appendix: output growth, inflation) [▶ details](#)
 - Result: Greenbook unemployment rate forecast error is **predictable** using either the first PC of all sentiments or the sentiment for 'economic activity' as predictor
- Therefore, Greenbook forecasts do **not** incorporate all information available to the FOMC
 - ⇒ there is additional information in the **text** that captures the Fed's information set
 - ⇒ should remove the contribution of the additional information to Δ FFR

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Comparison across Specifications

Table 1: R^2 across different specifications

		(1) R^2 with 10-word sentiment (main specification)	(2) R^2 with 5-word sentiment (robustness)
	Number of regressors		
Romer-Romer original OLS with subset of forecasts	19		0.50
Ridge with extended set of forecasts	133		0.55
Ridge with all forecasts & sentiments (linear)	429	0.65	0.66
Ridge with all forecasts & sentiments (nonlinear)	858	0.75	0.77
Ridge with all forecasts & sentiments (linear with lags)	1,613	0.87	0.88
Ridge with all forecasts & sentiments (nonlinear with lags)	3,226	0.94	0.95

- Preferred specification: 94% of FFR variation is systematic, **6%** explained by shocks
- R^2 for Romer-Romer OLS in original paper: 0.28

Comparison across Specifications

Table 2: R^2 across different specifications

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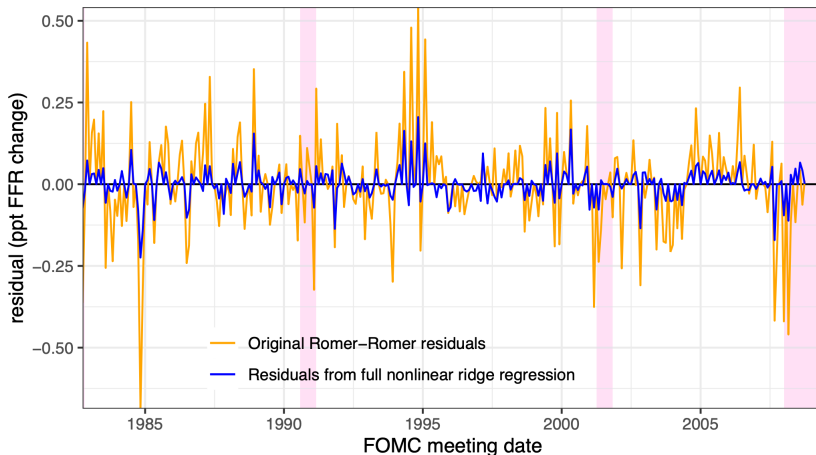
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Individual Regressors?

- Examining the model fit accounts for most of the (direct) model evaluation
 - With $\geq 3,000$ regressors, difficult to present an overview of the coefficient estimates
 - Hence, relative importance/contribution of individual regressors are unclear
- Paper stresses constructing an 'all-purpose' policy shock series as its goal
 - Construct a shock series that can be used to test the effect of policy on many different macro variables
 - Emphasis is put on predictive power rather than interpretation of coefficients

Estimated Monetary Policy Shocks

Figure 4: ESTIMATED MONETARY POLICY SHOCKS



What are Monetary Policy Shocks?

How can we interpret FFR target decisions that are 'nonsystematic/exogenous' ?

- FFR target decisions that are based on considerations *other than* the Fed's assessment of economic conditions → can be interpreted as “surprises” to the Fed staff
- Occur in nonsystematic ways (otherwise would have been captured by ridge regression)

Case study of largest shock episodes

- Nov 15, 1994 meeting: $\Delta FFR = 75 \text{ bps} = \underbrace{(54 \text{ bps})}_{\text{systematic}} + \underbrace{(21 \text{ bps})}_{\text{shock}}$
 - Staff forecast increase in output, suggest two policy options: 1) no change 2) $\uparrow 50 \text{ bps}$
 - Greenspan: “mild surprise would be of significant value” → proposed larger rate increase
 - FOMC participants agree, emphasize the *credibility* of controlling inflation

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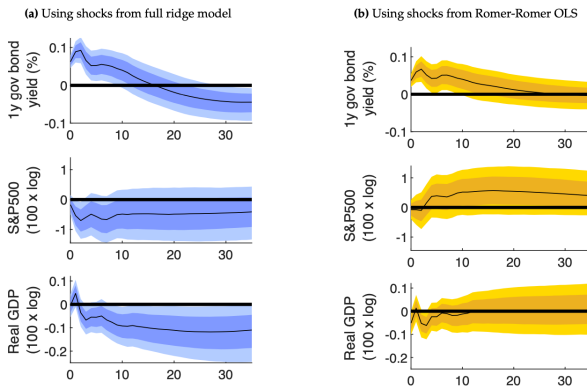
Having estimated policy shocks, now test how economic variables respond to the shocks

- Estimate a monthly **Bayesian VAR** following Jarocinski & Karadi (2020)
- Estimated shock series included as an exogenous variable ordered first in Cholesky ordering
- Variables included: 1-year Treasury yield, $\log(\text{S\&P 500})$, $\log(\text{real GDP})$, $\log(\text{GDP deflator})$, unemployment rate, excess bond premium (EBP) of Gilchrist and Zakrajsek (2012)
- Use shock series estimated for 1982:10 - 2008:10, but BVAR allows IRF estimation for full sample (-2016:12)

⇒ Compare impulse responses for ridge model & Romer-Romer OLS

Response to a Policy Tightening

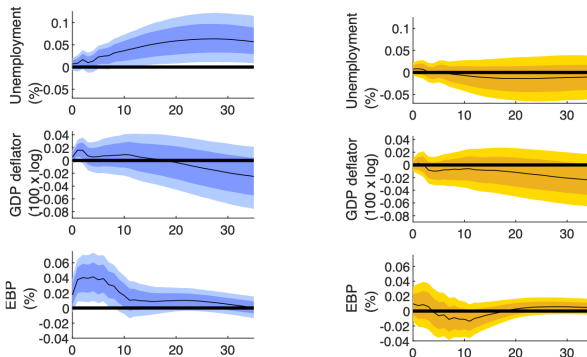
Figure 5: IRFS TO DIFFERENT MONETARY POLICY SHOCK MEASURES



Median, 16th/84th, 5th/95th pcs plotted. Based on 2000 draws from Gibbs sampler.

- Real GDP response almost flat under shock from Romer-Romer OLS

Response to a Policy Tightening



Median, 16th/84th, 5th/95th pcs plotted. Based on 2000 draws from Gibbs sampler.

- Unemployment rate and EBP response similarly insignificant under Romer-Romer OLS

Conclusion

- Presents a method for estimation of monetary policy shocks
 - Goal: remove variation in FFR that is due to information available to the FOMC
 - Utilizes textual content of documents prepared by Fed staff to capture intended changes in the policy instrument
- Resulting estimation: about 6% of FFR variation due to exogenous monetary policy shocks
- Estimated shocks lead to IRFs that are better in line with standard economic models

Comments

- **Sentiment indicators**

- For some concepts, not always clear what they are measuring - need judgment
 - e.g. Negative sentiment on 'inflation expectations' → likely related to *higher* inflation expectations in the future (as opposed to *lower*)
 - e.g. Negative sentiment on 'currencies' → ?
- Does not seem to be a problem for shock estimation. Including the full set of sentiment indicators seem to provide some notion of context to the regression
- In some cases, we might be interested in an economic interpretation of sentiment indicators → the set of concepts for which we can give an interpretation will be limited

- **Ridge regression**

- Tuning parameter λ chosen using k -fold cross validation with $k = 10$, but this choice of k seems somewhat arbitrary, no additional explanation (will also affect R^2)
- No explanation for considering nonlinearity

Appendix

Selection Algorithm [◀ Back](#)

1. Go through list of **triples** with ≥ 250 mentions (≈ 1 per meeting)
 $\Rightarrow$ select triples that are economic concepts
2. Go through list of **doubles** with ≥ 500 mentions
 $\Rightarrow$ select doubles that are economic concepts
3. If a selected double is a subset of one or more triples, **keep the triple(s)** if
 - The triples close to add up to the double & the triples are sufficiently different concepts (e.g. 'commercial real estate', 'residential real estate')
 - OR The double is too ambiguous by itself

Else: **keep the double** and unselect the triple(s)

4. Go through list of **singles** with ≥ 2000 mentions
5. If a selected single is a subset of one or more doubles, **keep the double(s)** if
 - The doubles close to add up to the single & the doubles are sufficiently different concepts
 - OR The single is too ambiguous by itself
(e.g. 'expenditures', 'government expenditures', 'capital expenditures', ...)

Else: **keep the single** and unselect the double(s)
(e.g. 'credit', 'consumer credit', 'bank credit', ...)

Examples of Positive/Negative Words

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Table 1: EXAMPLES OF WORDS ASSOCIATED WITH POSITIVE AND NEGATIVE SENTIMENT

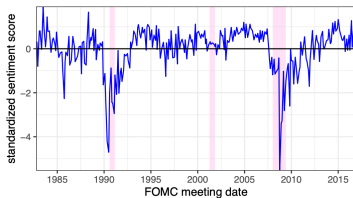
Positive sentiment	Negative sentiment
adequate	adversely
advantage	aggravate
benefit	bad
boost	burdensome
confident	collapse
conducive	concerning
desirable	decline
diligent	deficient
encouraging	eroded
excellent	exacerbate
...	...

Notes. Selected examples of words that are classified as expressing positive or negative sentiments in our augmented version of the dictionary of [Loughran and McDonald \(2011\)](#). The total number of classified words is 2,882.

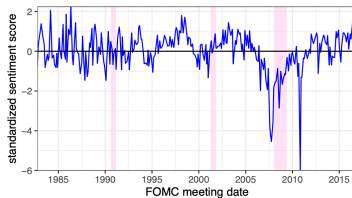
Sentiment Indicators (continued)

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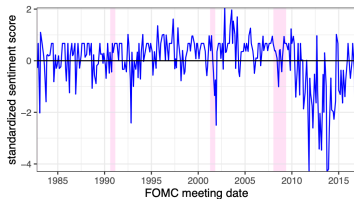
(e) Credit



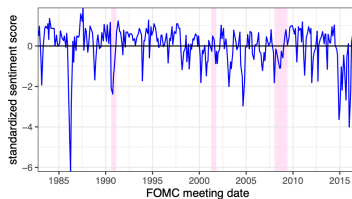
(f) Mortgages



(g) Fiscal policy



(h) Oil prices



- Ridge vs. LASSO regression
 - Penalty term uses the l^2 norm for ridge, l^1 norm for LASSO
 - Giannone, Lenza, Primiceri (2022): including many predictors can improve accuracy when predictors are **highly correlated**
 - Robustness test with LASSO regression (appendix)
- Optimal λ found using **k -fold cross validation**, with $k = 10$
 - Randomly divide the sample into 10 subsamples of equal size
 - For the $k = 1, 2, \dots, 10$ th subsample,
 - Fit the model excluding the k th subsample
 - Predict for the k th subsample and compute the mean squared error
 - Compute average MSE across the 10 MSEs
 - Repeat for different values of λ , choose the one that gives the smallest avr. MSE

What are Monetary Policy Shocks?

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- Nov 7, 1984 meeting: $\Delta FFR = -75 \text{ bps} = \underbrace{(-53 \text{ bps})}_{\text{systematic}} + \underbrace{(-22 \text{ bps})}_{\text{shock}}$
 - Fall in industrial production for the first time in 2 years
 - Smallest increase in employment since 1982
 - Increases in investment and consumption
 - Staff evaluates the slowdown as “pause in a recovery that has not run its full course”
 - Some FOMC members **disagree**, find staff forecast too optimistic
 - Large easing

Table 2: Greenbook Forecast Error Predictability Tests

	Dependent variable: Greenbook unemployment forecast errors							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	current quarter	1-quarter ahead	1-year ahead	2-years ahead	current quarter	1-quarter ahead	1-year ahead	2-years ahead
First PC of all sentiments	-0.029* [0.016]	-0.114** [0.049]	-0.445* [0.190]	-0.622** [0.238]				
Economic activity sentiment					-0.026 [0.016]	-0.098** [0.048]	-0.285* [0.165]	-0.363** [0.171]
Constant	-0.019 [0.014]	-0.070** [0.033]	-0.082 [0.121]	0.059 [0.201]	-0.019 [0.014]	-0.069** [0.035]	-0.077 [0.145]	0.160 [0.258]
R^2	0.045	0.149	0.248	0.208	0.033	0.097	0.090	0.055
Obs	210	210	210	62	210	210	210	62

- Forecast error = actual unemployment rate - forecast
- Interpretation: 1 s.d. $\uparrow$ in the sentiment PC associated with 0.445 pp negative forecast error

Differences in IRFs [◀ Back](#)

Unemployment rate: Sentiment indicators predict Greenbook unemployment rate errors [▶](#)

Stock prices:

- Fed systematically reacts to stock prices (Cieslak and Vissing-Jorgensen 2020)
- Imposing sign restrictions of Jarocinski and Karadi (2020) to distinguish between policy shock and information shock corrects the IRFs under Romer-Romer residuals

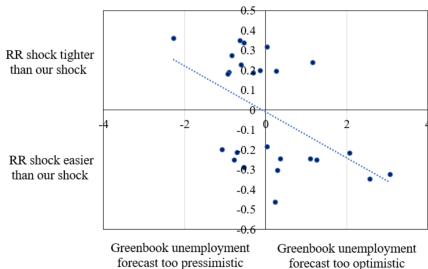
Credit spreads: Fed monitors changes in credit spreads - documents contain sentiment for 'spreads', 'credit standards', 'credit quality'

⇒ If Greenbook forecasts do not control for these systemic variations, resulting shock measure might reflect **spurious correlation** between FFR changes and economic variables

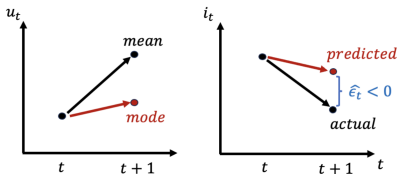
Greenbook Forecast Errors and Shocks

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(a) Relation between forecast errors and shock estimates



(b) Effect of too optimistic mode forecast on shock estimate



- (a): Scatter plot for 25 meetings with largest differences
- Predicting i_t only with *modal* forecast implies an *easing* shock
 $\Rightarrow$ **Easing shocks** are estimated when unemployment turns out **higher** than modal forecast